



Avg NDVI

0.311

Vegetation Health

Moisture Index

0.722

Soil Wetness Proxy

Healthy Fields



NDVI > 0.5

Stressed Fields



NDVI 0 – 0.2

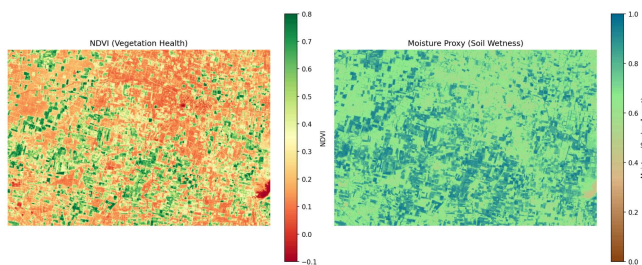


Irrigation Advisory

WATCH: Early Kharif season. Vegetation recovering. Adequate moisture detected. Monitor weekly.

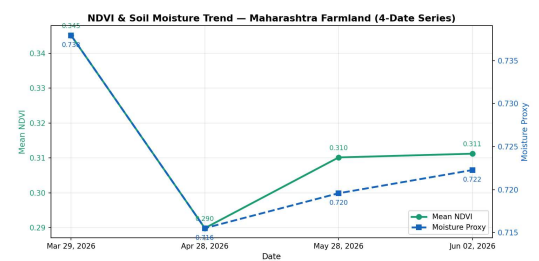
NDVI — VEGETATION HEALTH

Crop Health Analysis - Jun 02, 2026



Green = healthy vegetation | Red = bare/stressed

NDVI + MOISTURE TREND (4 DATES)

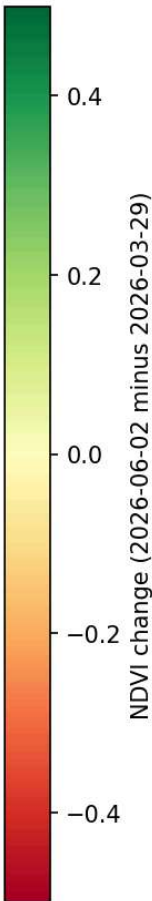
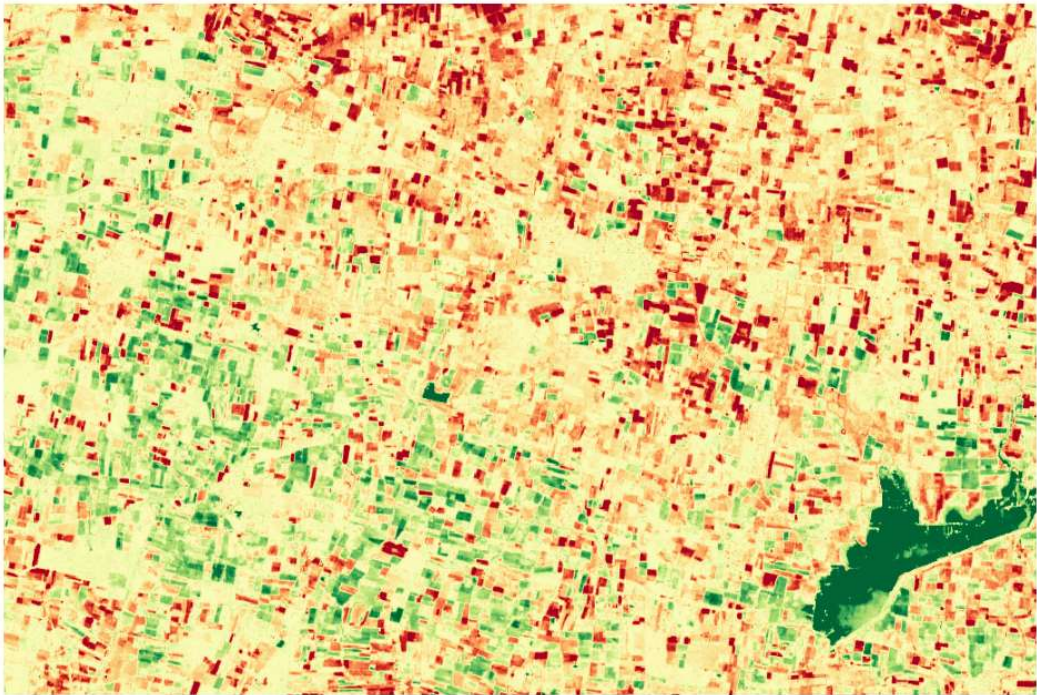


Green line = NDVI | Blue dashed = Moisture proxy

VEGETATION CHANGE MAP (MAR 29 → JUN 02)

45.8% decreased · 19.8% increased

Vegetation Change: 2026-03-29 -> 2026-06-02
(Green = increase, Red = decrease)



Green = vegetation increased | Red = vegetation decreased

FULL ADVISORY SUMMARY — ALL DATES

Irrigation Advisory Summary — All Dates

Date	NDVI	Moisture	Advisory
Mar 29, 2026	0.345	0.738	WATCH: Monitor closely. Consider irrigation if conditio...
Apr 28, 2026	0.290	0.716	WATCH: Monitor closely. Consider irrigation if conditio...
May 28, 2026	0.310	0.720	WATCH: Monitor closely. Consider irrigation if conditio...
Jun 02, 2026	0.311	0.722	WATCH: Monitor closely. Consider irrigation if conditio...

ABOUT THIS PROJECT



Data Source

Sentinel-2 L2A satellite imagery (ESA Copernicus). Bands B04 (Red) and B08 (NIR) downloaded for 4 dates: Mar 29, Apr 28, May 28, Jun 02 2026.



Methodology

$NDVI = (NIR - Red) / (NIR + Red)$ for vegetation health. Moisture Stress Index ($MSI = Red / NIR$) inverted as soil wetness proxy. Advisory logic combines both indices with growth-stage thresholds.



Irrigation Advisory Logic



Team Vikrama

NDVI < 0.25 + Moisture < 0.35 → Critical. NDVI < 0.35 + Moisture < 0.42 → Moderate stress. NDVI ≥ 0.35 + Moisture ≥ 0.42 → Healthy. Else → Watch.

Sanjivani College of Engineering, Kopargaon. Built for Agribusiness Hackathon 2026 (SPPU Research Park Foundation) and ISRO BAH 2026.